AI VIZARDS · AI & VISUALIZATION · JKU LINZ

Does a face tell CLIP your job?

Social bias in zero-shot occupation predictions on FairFace · Phase I: how does the model behave?

PAIR A · NO GENAI

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PAIR B · GENAI-AIDED

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Midpoint update · November 2026

THE QUESTION

A face carries no information about a job

So if CLIP’s occupation scores depend on gender, ethnicity or age, that systematic dependence is a bias.

Face
FairFace: ~108k faces, balanced over 7 ethnicity groups, labeled gender and age

→

CLIP ViT-B/32
matches it with “a photo of a nurse”, “... of an engineer”, ... zero-shot, no fine-tuning

→

Outputs only
one score per face and prompt; in Phase I, CLIP stays a black box

→

Compare groups
women vs. men, each ethnicity, each age group

FairFace labels are perceived attributes assigned by annotators; gender is binary in the dataset.

SETUP

Same question, two routes

Agreed as a group: CLIP ViT-B/32 zero-shot, the same 73,659 adult FairFace faces (tight crop), occupation prompts, compared across gender, ethnicity and age. Then each pair worked on its own.

	Pair A · no GenAI	Pair B · GenAI-aided
Who	Sonja Topf, Sebastian Kranzinger	Rareș Cazan, Diana-Alexandra Maxiniuc
Faces	73,659 adults (20+), train + validation	the same 73,659 adult faces
Prompts	43 jobs, “a photo of a/an …”	8 jobs, “a portrait photo of a/an …”
Measure	score vs. “a photo of a person”, group gaps	share of faces where a job is the top pick
Evidence	95% CI on every gap, reference check	bootstrap, subgroup control, regression
Views	interactive Plotly dot and box plots	seaborn heatmaps, bar and line charts

PAIR A · NO GENAI

Two checks before reading any gap

Score vs. person = an occupation's logit minus the same face's logit for "a photo of a person". A group gap of +1 means about 2.7× the odds.

1 · CLIP can read the labels

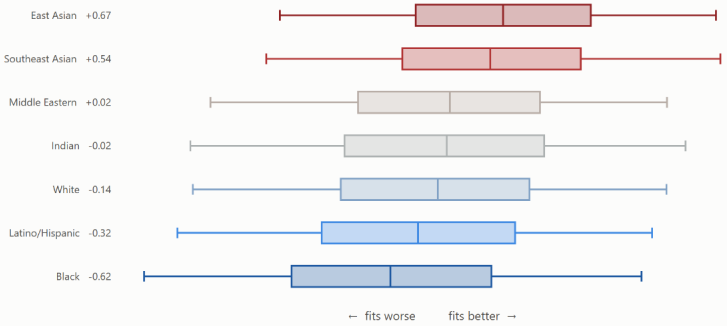
97% gender

63% ethnicity

43% age (4 words)

Share of faces where CLIP's zero-shot pick matches the FairFace label. Ethnicity mix-ups stay between neighbours: Middle Eastern → Latino 36%, White → Latino 33%, East Asian → Southeast Asian 27%.

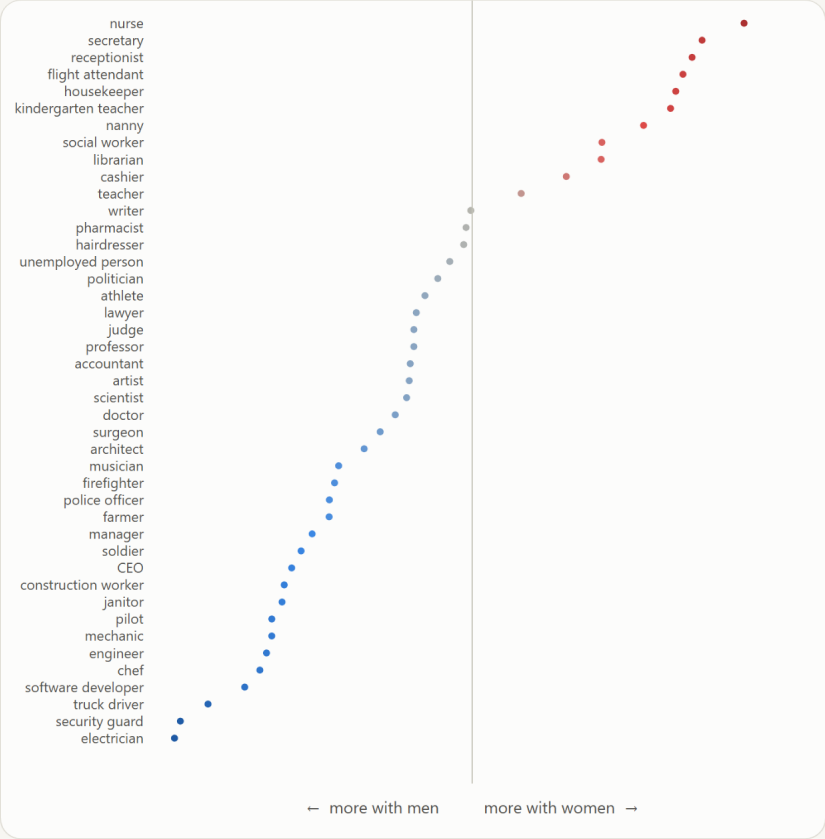
2 · The reference is not neutral



"A photo of a person" fits East Asian faces best (+0.67) and Black faces worst (−0.62). That shifts every score of a group, so we report it separately.

PAIR A · NO GENAI · LIVE DEMO

Almost every occupation is gendered



42 of 43 occupations differ significantly between women's and men's faces, and 31 of them lean male.

MORE WITH WOMEN

nurse +2.38 · secretary +2.01 · receptionist +1.92

MORE WITH MEN

electrician −2.60 · security guard −2.55 · truck driver −2.31

LIVE DEMO · VIEWS 2A–2C

Hover any dot to see its occupation, gap, rank and group sizes, for gender, ethnicity and age.

PAIR B · GENAI-AIDED

CLIP’s top pick among 8 jobs differs by group

40.1% vs. 6.5%

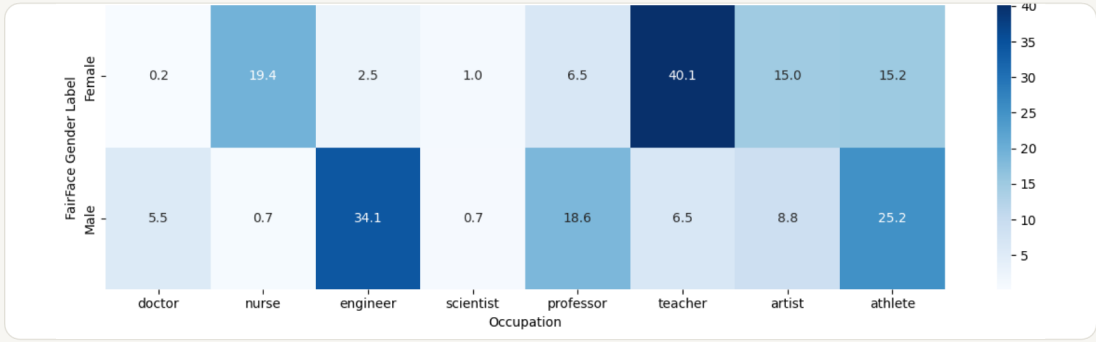
of women’s vs. men’s faces get “teacher”; “engineer” flips: 2.5% vs. 34.1%

40.1% vs. 8.6%

of Black vs. Indian faces get “athlete”; “engineer” flips: 6.7% vs. 37.2%

≤ 1.7%

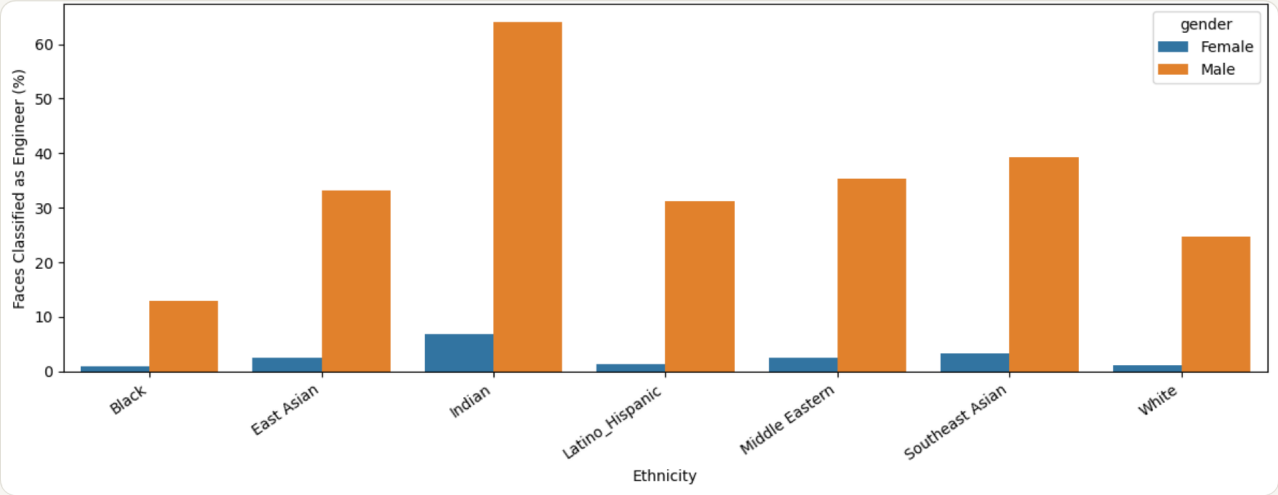
of any ethnicity group get “scientist”: the top pick shows only the winner, not how close the other jobs came





PAIR B · GENAI-AIDED

The gaps hold up under closer looks



“Engineer” as top pick by ethnicity and gender: not one fixed male effect. Indian men 64.1%, Black men 12.9%, women at most 6.8% in every group.

54.3% vs. 9.5%

“athlete” for Black vs. Indian men (women: 26.9% vs. 7.5%), and in every age group

31.5 points

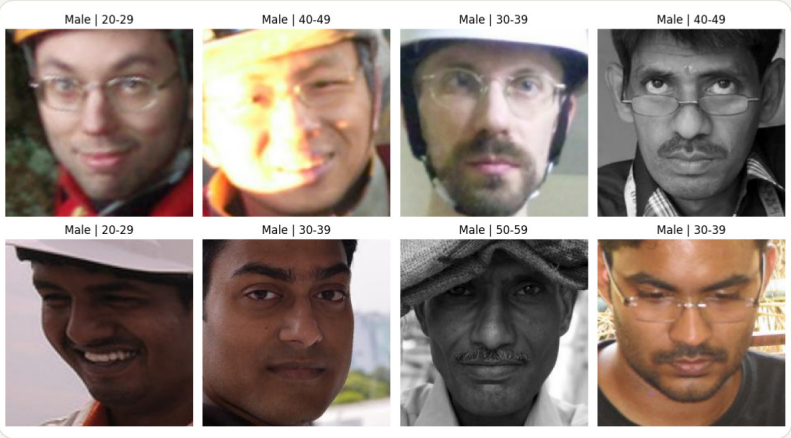
“engineer” gap, men minus women; 95% bootstrap interval 31.0 to 32.0

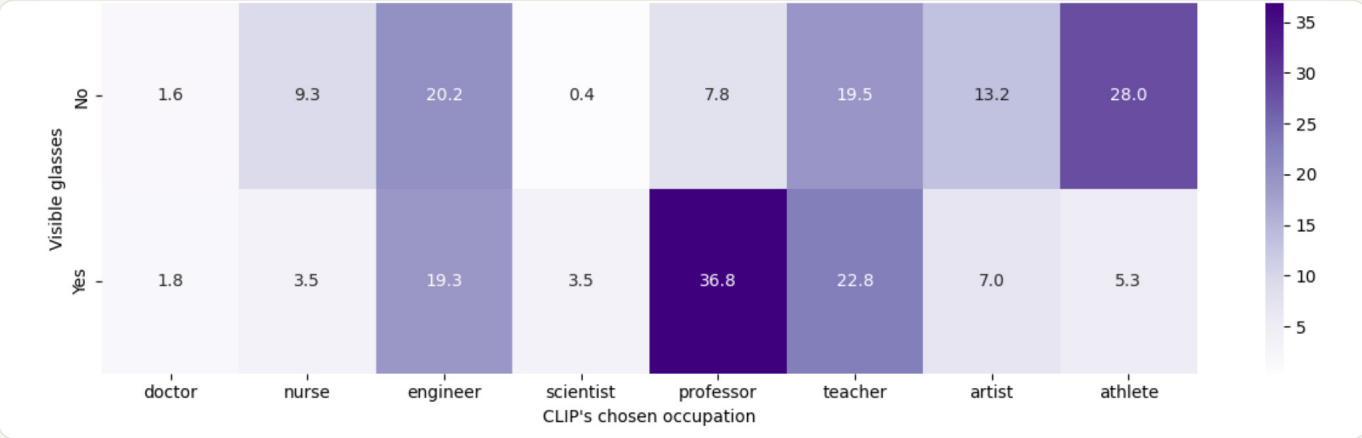
32.0 points

the same gap when comparing only faces of the same age and ethnicity (42 subgroups)

PAIR B · GENAI-AIDED

A clue in the images: glasses





57 of 314

clearly labeled adult faces wear glasses (one unsure left out), labeled by hand in an AI-built widget

36.8% vs. 7.8%

get “professor” with vs. without glasses; “athlete” drops from 28.0% to 5.3%

Holds with controls

professor, scientist, teacher and engineer scores stay higher after adjusting for gender, age and ethnicity

PAIR B · GENAI-AIDED

The answer depends on how you ask

42–46%

of faces get a different top job when only the wording changes

“a portrait photo of a …” vs. “a photo of a person whose job is a …” or “… of someone working as a …”. Close calls flip most: ≈63% vs. 14–22% for clear wins.

32 → 6

points: the engineer gender gap with “a photo of someone working as a …”

Men’s “engineer” falls from 34.1% to 6.3%, and “teacher” becomes the top pick for both genders (63% of women, 38% of men).

40 → 60%

of women’s faces get “teacher” once nurse and athlete leave the list

The teacher gap between women and men grows from 34 to 51 points.

COMPARISON

Different measures, same core associations

Gender rows are women vs. men. Both pairs score the same 73,659 faces. Pair A: logit gaps over 43 jobs. Pair B: % of faces, top pick among 8 jobs.

Association	Pair A · score gap	Pair B · top-pick share	Verdict
Women ↔ nurse	+2.38, largest of 43 gaps	19.4% vs. 0.7%	Agree
Men ↔ engineer	−1.80	2.5% vs. 34.1%	Agree
Black ↔ athlete	+0.64, top job for Black faces	40.1% vs. 8.6% (Indian)	Agree
Indian ↔ engineer	+0.63, 2nd for Indian faces	37.2% vs. 6.7% (Black)	Agree
Women ↔ teacher	+0.43, only rank 11 of 43	40.1% vs. 6.5%, largest gender gap	Differs

Why “teacher” differs

With 8 options, “teacher” wins whenever the others fit worse, and Pair B showed that share moves with wording and list. Pair A’s per-job score avoids that competition but depends on the reference prompt. Each pair exposed a blind spot in the other’s measure.

AI4VIS

What changed with GenAI in the loop

Breadth

Pair B ran about ten experiments, from a labeling widget to regression and a PCA of the scores. Pair A ran fewer views, with a 95% CI on every gap.

Visual choices

Pair B: default seaborn heatmaps and bars, quick and static. Pair A: hand-designed interactive dot plots, sorted, with diverging colors and hover details.

Speed vs. verification

AI turned questions into working code fast, but the code needed fixes and its interpretations had to be checked. When brainstorming, it got stuck on literal wording rather than meaning.

Who steered

Ideas stayed human: the glasses hypothesis came from looking at faces, and each follow-up came from a result. AI wrote the code and drafted interpretations that Pair B checked.